Cut-regime models — O_2 carbon steel

The equations behind the cut-settings suggestions, with each symbol explicit. These are **starting models, not guarantees**. Most published experiments are flat plate and CO_2; a beam section adds access effects (recessed web, far wall, 5-axis head) that act as a separate multiplier.

Companion files:

- `docs/cut-regime-data-spec.md` — definitions, outcome codes, and the data row schema.
- `docs/cut-regime-data-sheets.md` — printable collection sheets.
- `data/cut-regime/runs.csv` — the run log (one row per sample).
- §12 — proposed map views.
- §15 — verification log (what was checked against which source, and what changed).

Status: **v0.4** (2026-11-09). Changes from v0.3 are in §13.

Trust tiers

- **T1 — exact / verified:** unit conversions, energy accounting, optics, machine facts.
- **T2 — supported / semi-empirical:** checked against a source, but the source is CO_2, flat plate, a different machine, or a different definition. Calibrate the numbers.
- **T3 — hypothesis / fit template / shop extrapolation:** useful, not laws.

Regimes

The practical picture is a few regimes in $(speed, pressure, focus, power)$ space:

- **N0 no penetration** — not enough energy to sever.
- **N1 through, bridged** — full kerf, still joined by melt or slag.
- **N2 forced separation** — separates only with force, tap, or hammer.
- **N3 free separation** — releases with no tool. This is the usable outcome.
- **Edge quality** (for N2/N3 only): striation (S0–S3, per zone), mirror, dross (D0–D3).

Outcome codes and the handling test are in `docs/cut-regime-data-spec.md` §5.

Read this first

$$S = 0.06\,\frac{P}{v\,t} \qquad [\mathrm{J/mm^2}]$$

P = beam power in W · v = speed in m/min · t = thickness in mm · S = beam energy per unit **cut-face area** (wall or flange area, t times length) in J/mm².

The 0.06 is exact: 1 m/min = 1000/60 mm/s, so $\frac{P}{v \cdot \frac{1000}{60}\,t} = 0.06\,\frac{P}{v\,t}$. (T1)

Worked example — 30% of 12 kW, 3.2 m/min, 5 mm web (assumes linear % → kW, unverified):

$$P = 0.30 \times 12000 = 3600\ \mathrm{W}$$

$$S = \frac{0.06 \times 3600}{3.2 \times 5} = 13.5\ \mathrm{J/mm^2}$$

What S is and is not. It is the beam energy per cut-face area. It is *not* the total energy the process needs. Oxidation adds energy (§2), so the total is larger than S. The name "severance energy" is ours; we have not confirmed it as a literature term.

Which energy?

- S_{beam} — the default. Uses beam power P. This is what we measure.
- S_{eff} — uses an absorbed-plus-oxidation energy (§2). Only when the oxidation term is estimated.

Every table, plot and comparison states which one is used. Do not compare literature values (often S_{eff}, flat plate, or CO_2) against shop S_{beam} without saying so.

Symbols

Symbol	Meaning	Unit
P	beam power (set/measured at the workpiece)	W
v	cutting speed	m/min (shop) or mm/s (SI)
t	material thickness (wall, flange or web)	mm
S	beam energy per unit cut-face area	J/mm²
p	assist-gas (O_2) pressure at the cutting head	bar (gauge or absolute: **to confirm**)
f	focus offset from the surface (positive = above)	mm
d_f	focused spot diameter (1/e² or D4σ — state which)	mm
r_g	focused spot radius, $d_f/2$	mm
z	distance from the beam waist along the axis	mm
z_R	Rayleigh range (depth of focus)	mm
λ	laser wavelength (fibre ≈ 1.07 µm)	µm
M^2	beam-quality factor (1 = perfect)	–
I	mean intensity over the spot ($4P/\pi d^2$)	W/mm²
η_{abs}	absorptance: fraction of beam power absorbed by the cut front	–
f_{ox}	fraction of the removed iron that is oxidised to FeO	–
ΔH_{ox}	heat of Fe → FeO oxidation, ≈ 250 kJ/mol	J/mol
P_{ox}	power released by iron oxidation = $f_{ox} \cdot \dot{m}_F e / M_{Fe} \cdot \Delta H_{ox}$	W
$X_{\mathrm{O_2}}$	oxygen purity (0–1)	–
w_k	kerf width	mm
α	cut-front / taper angle	°
ρ_g	assist-gas density	kg/m³
v_g	assist-gas velocity at the kerf	m/s
δ	melt-film thickness	mm (not measured)
γ	molten-steel surface tension	N/m
κ	thermal diffusivity of steel (≈ 5.9 mm²/s, T2 — from a published iron value)	mm²/s
h	nozzle standoff	mm
d_n	nozzle diameter	mm
Pe	Péclet number, moving Gaussian source: $v \cdot r_g/(2\kappa)$	–

Symbol	Meaning	Unit
We	Weber number (gas drag ÷ surface tension)	–

Symbol clash fixed in v0.3: v0.2 used η as "coupling efficiency" and in the oxidation sum. The literature uses η for different things (in one cutting model it is the *oxidised fraction*). v0.3 splits them into $\eta_a bs$ and f_{ox}.

Shapes at a glance

Plot	Shape	What it tells you
v vs t at fixed P	hyperbola	thicker → slower
v vs P at fixed t	straight line through 0	power buys speed proportionally
$logv$ vs $logt$	straight line, slope –1	the same-energy line
S vs p	island / lens	mirror window, bounded by striation and dross (T3)
focus vs quality	parabola	best focus near minimum roughness/dross (T3)

Worked examples from the shop

Cut	Power	Speed	t	S_{beam}	Outcome
150UB14 web, conservative	25% = 3000 W	2.0 m/min	5 mm	**18.0**	good (shop label)
150UB14 web, mirror (slightly unstable)	30% = 3600 W	3.2 m/min	5 mm	**13.5**	mirror, unstable (shop label)
6 mm SHS, primed	25% = 3000 W	2.0 m/min	6 mm	**15.0**	good (shop label)
10% test cut	10% = 1200 W	1.2 m/min	6 mm	**10.0**	to be graded

These are **shop labels, not graded outcomes.** They become data only once re-cut and graded with the spec. The 13.5 cut shows that S alone does not separate mirror from unstable.

1. Energy and the penetration boundary

$$S = 0.06\,\frac{P}{v\,t} \qquad [\mathrm{J/mm^2}]$$

$$v = 0.06\,\frac{P}{S\,t} \qquad [\mathrm{m/min}]$$

$$P = \frac{S\,v\,t}{0.06} \qquad [\mathrm{W}]$$

No-cut boundary: the cut needs $S \geq S_{\min}(t)$:

$$v \leq 0.06\,\frac{P}{S_{\min}(t)\,t}$$

Shape: a hyperbola in (v, t); a straight line through the origin in (v, P) at fixed t; a straight line of slope –1 in $logv$ vs $logt$. These follow from the algebra (T1).

Fitting: $S_{\min}(t)$ is empirical. Fit linear or power law (see §9).

2. Oxidation energy (oxygen reaction)

The iron–oxygen reaction adds heat. Powell et al. (2009) give the Fe → FeO enthalpy as **≈ 250 kJ/mol** between the melting and boiling points of iron, and about 600 kJ/mol above the boiling point (T1 for the number; T2 for

applying it to a cut).

Rate of iron removed from the kerf, and the power from oxidising a fraction f_{ox} of it:

$$\dot{m}_{\mathrm{Fe}} = \rho_{\mathrm{Fe}}\, v\, t\, w_k$$

$$P_{\mathrm{ox}} = f_{\mathrm{ox}}\, \frac{\dot{m}_{\mathrm{Fe}}}{M_{\mathrm{Fe}}}\, \Delta H_{\mathrm{ox}}$$

$$P_{\mathrm{eff}} = \eta_{\mathrm{abs}}\, P + P_{\mathrm{ox}}$$

$$S_{\mathrm{eff}} = 0.06\, \frac{P_{\mathrm{eff}}}{v\, t}$$

What the sources say, and what they do not:

- Powell (2009, CO_2): in a typical sample, *about half the iron in the cut zone* was oxidised to FeO. This is a fraction of **iron**, not of **energy**. The melt that leaves is an Fe/FeO mixture.
- A secondary source (KEM 442.81, 2010, citing other authors' models) puts the combustion contribution at **40–70%**, and one of those models used η = 50% **as the oxidised fraction**. So the "40–60%" figure in v0.2 mixed two quantities. Restated: the reported share of energy is roughly 40–70%, depending on source, speed and kerf (T2).
- Our estimate (5 mm, 25% = 3000 W, f_{ox} = 0.5, $\rho_F e$ and ΔH from Powell, assumed kerf):

v (m/min)	kerf (mm)	P_ox (W)	P_ox ÷ P	S_{beam}	S_{eff} (η_abs = 1)
2.0	0.2	588	0.20	18.0	21.5
2.0	0.3	881	0.29	18.0	23.3
2.8	0.3	1234	0.41	12.9	18.1
3.6	0.3	1586	0.53	10.0	15.3

Read this as: oxidation is a **significant, speed-dependent** term, and its share moves with kerf width and f_{ox}, both of which we do not measure. A fixed "40–60%" is not a constant. The kerf and f_{ox} values here are assumptions, so the table is an illustration, not a result (T2).

Shape: the apparent S_{min} falls as the oxidation share grows with thickness and oxygen supply.

3. Power density and focus

$$I = \frac{4P}{\pi d_f^2} \qquad [\mathrm{W/mm^2}]$$

$$d(z) = d_f \sqrt{1 + \left(\frac{z - z_0}{z_R}\right)^2}$$

$$z_R = \frac{\pi w_0^2}{M^2 \lambda} = \frac{\pi d_f^2}{4 M^2 \lambda}, \qquad w_0 = \frac{d_f}{2}$$

- **Verified (T1):** the Rayleigh range with M² and the waist radius is the standard Gaussian result (RP Photonics; Kogelnik & Li 1966; ISO 11146 for M²). The $\pi \cdot d_f^2/(4 \cdot M^2 \cdot \lambda)$ form is exact when d_f is the **waist diameter** and λ is in the same unit.
- **Correction (v0.3):** $I = 4P/(\pi d^2)$ is the **mean** intensity over the 1/e² diameter. The **peak** is $I_0 = 8P/(\pi d^2)$, twice as large. Label every plot with which one is used.
- Diameter definition matters: 1/e² and D4σ agree for an ideal Gaussian and disagree for real beams. State which we use.

Use the **normalised focus** f/d_f rather than raw millimetres when comparing optics.

Shape: I peaks at the waist and falls as a hyperbola either side. Quality against focus is a parabola with a minimum near the optimum (T3).

4. Striation-free / mirror boundary (semi-empirical, T3)

No universal closed form. The usual fitted form is a power law:

$$v_{\text{crit}} = C \left(\frac{P}{t}\right)^{a} p^{b} \left(\frac{f}{d_f}\right)^{c}$$

$$\log v_{\text{crit}} = \log C + a \log \frac{P}{t} + b \log p + c \log \frac{f}{d_f}$$

- Context (T2): a review of thin-sheet mild steel O_2 cutting reports striation appearing at low speed and disappearing ("steady cutting") above a critical speed, quoted as about 2 m/min for thin sections. Our thick-section speeds (2–4 m/min) sit near or above that, so the trend is consistent. It is not a calibration for our 5–25 mm sections.
- The **mirror window** is $v_{\text{crit}} \le v \le v_{\text{dross}}$. Below v_{crit}, striation. Above v_{dross}, dross. On an (S, p) plot this is a **closed island** if both bounds are real.
- **Source check:** the primary paper, Sobih, Crouse & Li (2007), was **not accessible** (bot check). Its critical-speed model is unverified. Its scope is EN43, **1–2 mm**, **1 kW fibre**. It is not a source for 5–25 mm sections at 12 kW.

5. Dross / ejection boundary (semi-empirical, T2/T3)

$$We = \frac{\rho_g \, v_g^2 \, \delta}{\gamma}$$

δ is not measured, so `We` cannot be computed from our logs. Use p and v as observable proxies and treat `We` as a hypothesis.

Dross-free cutting needs enough gas momentum and pressure. Too much oxygen over-drives the oxidation and causes striation and side burning. The boundary is commonly fitted as:

$$v_{\text{dross}} = C \left(\frac{P}{t}\right)^{a} p^{b} \qquad (\text{T3 fit template})$$

$$p_{\min}(v, t) \;\ldots\; p_{\max}(v, t)$$

Pressure reference (T2, checked):

Source	Process	Oxygen pressure
Powell et al. (2009)	CO_2, head supply	≈ 1.0 barg for > 3 mm; up to ≈ 5.0 barg for thinner
Genna et al. (2020)	CO_2, O_2 assist	≈ 0.2–0.6 bar, depending on thickness
Laser Focus World (2010)	fibre, 15 mm mild steel	1.0 bar O_2, 1.5 mm nozzle, focus 7 mm below surface
Shop (`cutParams.ts`)	12 kW fibre, 3–25 mm	0.6–0.7 bar

Our 0.6–0.7 bar is inside the wide literature range, but it is **below Powell's ≈ 1 barg for thick sections**. The shop value is also ambiguous: gauge or absolute is not stated. Confirm which, and measure it at the head (see §11).

Shape: a lens or ellipse in (v, p): dross at low speed or low pressure, and again at excessive speed.

6. Oxygen purity

$$v \propto X_{\mathrm{O_2}}^{n} \qquad (\mathrm{T3\ fit\ template})$$

What the source says: Powell et al. (2009), citing Ivarson et al. (1993) and Powell et al. (1992), report that **cut speed falls by 50% or more when O_2 purity drops by 3%**. Their data are CO_2. (The v0.2 claim of "1.25% impurity halved speed" was **not supported** by this source and is removed.) I have not found a fibre-laser purity curve, so the fibre sensitivity is unknown.

Our gas (Supagas, verified from the Supagas oxygen technical sheet, 2018 version):

Grade	Typical purity	Notes
Industrial	> 99.5%	Diving-grade listed at the same level
HP	> 99.8%	< 10 ppm moisture
UHP	> 99.999%	not needed for cutting

- **12-pack:** 124.0 m³ at 101.325 kPa, 15 °C; 600 L water capacity per pack (Supagas sheet). Pack pressure is not on the sheet we read. From 124 m³ and 600 L, the fill pressure is roughly 200 bar absolute (T2 estimate, not from the sheet).
- **Run-time per pack:** $124 m^3 \div Q$, where Q is the flow at the head. Q depends on pressure and nozzle, and we have not measured it. Measure Q (or the pack-pressure drop per shift) before quoting run-time.
- **Grade purchased: Industrial, >99.5% (decided, shop policy: cheapest grade).** Nominal X_{O2} = 0.995 for every row unless the batch certificate says otherwise.
- **Purity is not a test variable for us.** We cannot buy HP, so we do not need an HP-vs-industrial series. Record the certified purity per pack anyway (`gas_purity_pct`), because a bad batch changes speed.
- **Limit of our knowledge:** the –3% → –50% speed figure is at about 97% purity. We have no data at 99.5%, so the size of the purity effect in our range is unmeasured.

7. Kerf

$$w_k \approx d_f + 2\,t\,\tan\alpha$$

α is the taper angle. It changes with speed, focus and pressure, and can change sign as the focus passes through the thickness. The formula is a geometric approximation (T2).

8. Dimensionless groups

Péclet number (corrected in v0.3). For a moving Gaussian source, the standard form uses the beam radius (Manca et al. 2007 COMSOL proceedings; arXiv 2202.06501):

$$\mathrm{Pe} = \frac{v\,r_g}{2\kappa}, \qquad r_g = \frac{d_f}{2}$$

v0.2 used $v \cdot d_f / \kappa$ (wrong factor and sometimes the wrong length). v0.2's $v \cdot t / \kappa$ is **not** the standard definition. Thickness belongs in a separate ratio, the aspect ratio t / w_k, if we use one.

Worked check (T2): with κ ≈ 5.9 mm²/s and a 0.2 mm spot (r_g = 0.1 mm):

v (m/min)	Pe
2.0	0.28
4.5	0.64

So in our speed range, Pe is below about 1 for a 0.2 mm spot. Whether Pe is the right collapse variable for cutting is a hypothesis (T3). Test it, do not assume it.

$$\Phi\left(S,\ \mathrm{Pe},\ \mathrm{We},\ \tfrac{f}{d_f},\ \tfrac{h}{d_n},\ X_{\mathrm{O_2}}\right) = 0 \qquad (\mathrm{T3\ hypothesis})$$

$$\mathrm{We} = \frac{\rho_g\, v_g^2\, \delta}{\gamma} \qquad (\delta\ \mathrm{unmeasured})$$

If the groups match, the shape of the map should repeat across sections and machines (T3). The coefficients still move with optics, nozzle and material.

9. Fitting recipe

1. Fix optics, nozzle, gas and material. Record them on every row.
2. Log each sample as per `docs/cut-regime-data-spec.md`: section, feature, face, t, P, v, f, p (with gauge/absolute), h, d_n, gas grade and purity, S_{beam}, outcome codes.
3. Outcome codes: `through_code` N0–N3, plus edge grades (`striation` per zone, `mirror_flag`, `dross_grade`). Do not use one "mirror / striated / dross" label.
4. Penetration: fit $S_{\min}(t)$ from N0 vs N1 (linear or power law).
5. Separation: N2 vs N3 gives the separation boundary, separate from penetration.
6. Mirror: logistic regression on $logS$, $logp$, f/d_f for the mirror flag. The 0.5 contour is the boundary. Needs about 30+ points per class, not 10–15.
7. Dross: fit upper and lower branches in (v, p).
8. Do not extrapolate outside the tested window.

10. Trust tiers and fact-check

Item	Tier	Status / note
$S = 0.06 \cdot P/(v \cdot t)$ unit conversion	T1	Exact: 1 m/min = 1000/60 mm/s.
Rayleigh $z_R = \pi w_0^2/(M^2\lambda)$; `d(z)`	T1	Standard Gaussian result; RP Photonics, Kogelnik & Li 1966.
$I = 4P/(\pi d^2)$	T1	Mean over 1/e² diameter; **peak is 2×**. Label which.
Penetration / no-cut boundary	T1	Energy balance; only $S_{\min}(t)$ is empirical.
Fe → FeO enthalpy ≈ 250 kJ/mol	T1	Powell et al. (2009), Guideline 5.
Oxidation share of energy 40–70%	T2	Secondary citation (KEM 442.81). Not one number; kerf and f_{ox} dependent.
"Half the iron oxidised"	T2	Powell (2009), CO_2 typical sample. Fraction of iron, not energy.
Purity: –3% → speed –50%	T2	Powell (2009), citing 1992/1993 work. CO_2.
Pe = $v \cdot r_g/(2\kappa)$	T2	Moving Gaussian-source definition (Manca 2007; arXiv 2202.06501). Not cutting-specific.
Pe with thickness (v0.2 $v \cdot t/\kappa$)	**removed**	Not a standard definition.
`We` (dross)	T2	Standard Weber number; δ unmeasured.
Striation-free law and v_{crit}	T3	Sobih et al. (2007) not accessible; scope 1–2 mm, 1 kW.
v_{dross} power law	T3	Generic fit; calibrate on the machine.
Dimensionless collapse $\Phi = 0$	T3	Hypothesis.

Item	Tier	Status / note
S = 4–13 J/mm² for MS + O_2	T2	Published range, not verified against a primary source.
Shop good cuts 13–18 J/mm²	T3	Small local dataset, not graded.
Focus sign: `cutParams.ts` table is `+` (above surface); shop practice is **below** surface	**open, likely table error**	Practice and one fibre source (`−7` below, 15 mm) agree. See §11.
O_2 pressure 0.6–0.7 bar	T2	In literature range; gauge/absolute unknown; below Powell's thick-section value.
Purity sensitivity	T2	See above.
Nozzle / surface rules	T1 (shop)	Shop practice, not literature.
$r/min = v/(2\pi R)$	T2	Formula exact; R depends on the calibrated B-axis and eccentric.
"Flat plate" caveat	T1	Confirmed scope of the published maps.
Calibrate	T1	Must be measured; non-linearity shifts every S.
Mirror Rz threshold	TBD	Set from the calibration set (spec §11), not from literature.

11. Beam-section caveats and machine facts

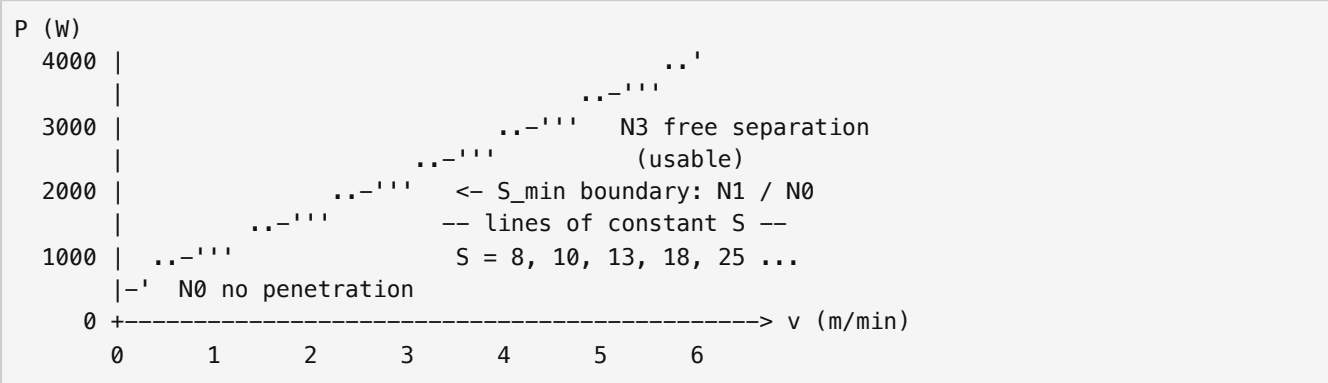
- The published maps are **flat plate and mostly CO_2**. A beam web is a recessed channel with different gas escape and height-sensor behaviour. Fibre and CO_2 may differ in absorption and in the oxidation share; we have not verified the size of that difference.
- Treat **external wall** and **recessed web** as separate series (an access factor on S or on `We`), not one curve.
- **Focus sign (shop practice confirmed by the user, v0.4).** The operator runs focus **below the top surface**, inside the material or past it. The data spec uses physical sign: **negative = below the top surface**, positive = above. Under that convention the practice is negative focus. The table in `src/lib/machine/cutParams.ts` stores **positive** focus (+4 to +12, "above the surface"), and `profile.ts` sets `focus.sign = −1` so the CAM receives negative values. Both conflict with the practice above, and with the fibre source that puts 15 mm mild steel at −7 mm. Likely cause: the table's sign is wrong, not the practice. **Do not change the code until a focus-sign test confirms it on the machine** (see sheet 0, focus sign check).
- **Machine documentation (corrected in v0.4).** The repo doc `docs/machine-kl90010t.md` is detailed on the **envelope** (length, diameter, thickness, bevel, nesting limits), and I verified those against the 8 Star page. It does **not** list laser source power, gas pressure range, nozzle list, or focus convention. Those figures are in `profile.ts` and `cutParams.ts` as code comments and table values, with no document behind them. The linked TubePro control manual (FSCUT, 66 pp, read) defines "Cut focus" (focus location in cutting) and "Cut PWR" (peak power) but gives no sign convention and no pressure units. Its gas page reads "actual pressure" from a DA output, so pressure needs a DA-to-bar calibration. The repo's controller name is Weihong WiseCAM/NcStudio, and the manual is FSCUT TubePro. **Confirm which controller is on the machine. Action:** photograph the laser source nameplate and the controller model plate.
- Calibrate the controller's . If it is non-linear, every S value shifts.
- The rotary axis is the calibrated B-axis centre, not necessarily the section centroid. The eccentric offset changes the radius used for r/min.
- **Corners and curves are separate feature classes** (spec §8).
- **Pressure unit (v0.4).** The shop states pressure in **bar**. Gauge or absolute is still not stated, and the TubePro gas page reads actual pressure from a DA output, so calibrate DA → bar at the head before using any value in a map.

12. Map views (proposed)

All maps are **planned, not built.** They need data first. Each map is for one series and states its S basis and its pressure basis (gauge or absolute).

12.1 Operating map — v vs P at fixed t (primary)

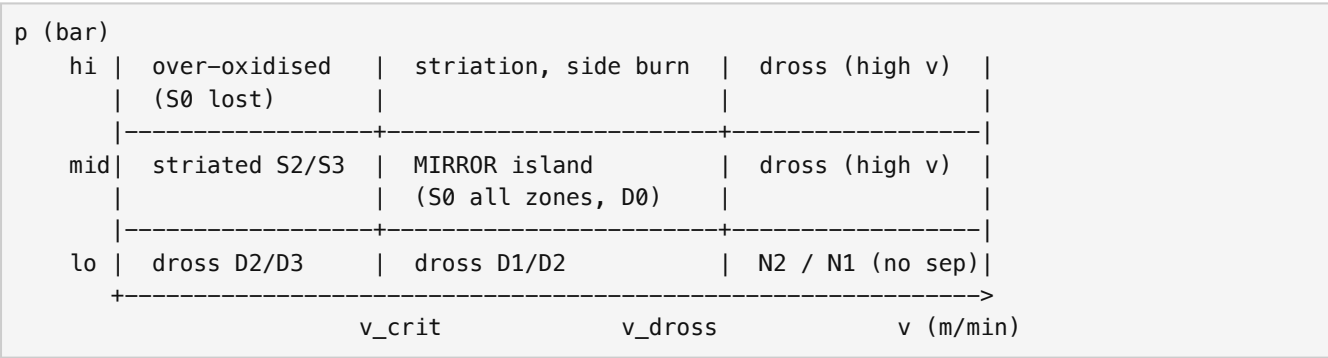
The map an operator uses. Lines of constant S are straight lines through the origin.



- Hard boundary: the N0/N1 line through the origin (T1 energy balance, with S_{min} fitted).
- Second boundary: the N3/N2 line, same shape, higher S.
- Overlay quality classes as markers (S0–S3, dross, mirror).
- Operator view: pick a constant-S line and move along it.

12.2 Quality map — v vs p at fixed P and t (T3 shape)

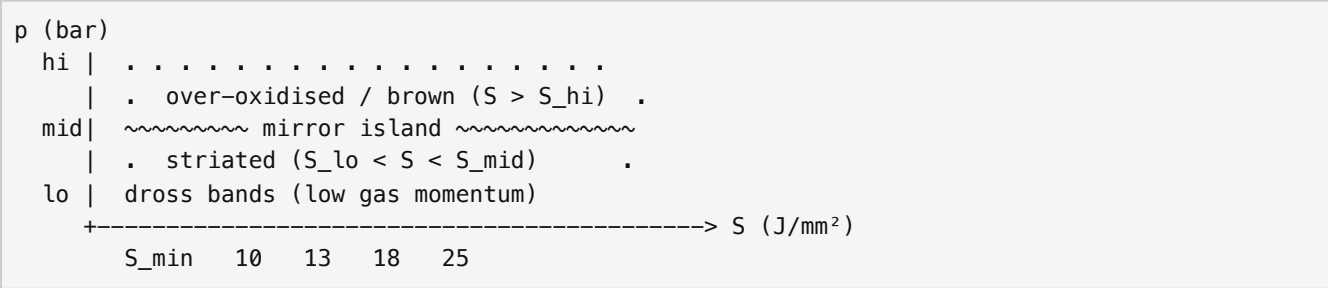
Shows the mirror island and its limits. Needs the most data.



- Every cell label is a hypothesis (T3) until graded data sits in it.
- The mirror cell is only defined when `S0` holds in all three zones and `Rz` is under the calibrated threshold.

12.3 Contour map — quality as contour bands in (S, p)

Collapses P, v, t into S. Tests the "one number" claim.



- Test whether the mirror island collapses onto one S band across thicknesses. If not, oxidation share, `Pe` or t matter separately.
- Only valid if S is S_{beam} for the whole plot.

12.4 Zone map — quality by depth

One panel per series, edge quality from top to bottom of the wall.

```
depth        S0  S1  S2  S3    Rz (um)   dross
top   ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓   ...
mid   ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓   ...
bot   ▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓▓   ...
```

- Shows where striation starts and how it develops. A cut that is mirror only at the top is not a mirror cut.
- Small multiples: one panel per (v, P) setting, so the operator sees how zones change along a speed line.

12.5 Focus and pressure — small multiples (T3)

Focus against quality, one panel per pressure. The parabola's minimum should move with t. Needs focus varied in the series, which the pilot does not do.

12.6 Feature and access split

Separate maps for each feature class (`straight`, `curve`, `corner`) and face (`web`, `flange`, `wall`, `kept/scrap`). Do not overlay corner or curve points on straight-line maps.

12.7 Build order

1. **12.1 operating map** — needs only N0/N1/N3 and S.
2. **12.4 zone map** — needs three-zone striation grades (the spec records these).
3. **12.2 quality map** — needs about 30+ points per class and a calibrated mirror threshold.
4. **12.3 contour test** — only after 12.1 and 12.2 exist.

13. Change log

Version	Change
v0.1	Initial draft. Mirror/striated/dross labels; Pe with d_f; S defined per kerf area.
v0.2	Outcome codes N0–N3 and edge grades. S as per cut-face area. Pe changed to $v \cdot t/\kappa$. S_{beam} vs S_{eff} rule. Map views added.
v0.4	Gas grade fixed to Industrial (>99.5%, shop policy). Purity removed as a test variable. Focus practice recorded as below surface (negative); `cutParams.ts` sign flagged as likely wrong. Pressure stated in bar, gauge/absolute open. Machine documentation status corrected (envelope documented; power, gas and focus not). TubePro manual checked.
v0.3	**Source check.** Pe restored to the moving-source form $v \cdot r_g/(2\kappa)$; v0.2 $v \cdot t/\kappa$ removed. Purity claim changed from "1.25% halved speed" to Powell's "–3% → –50%" (CO_2). Oxidation section rewritten: "half the iron" ≠ "40–60% of energy"; ΔH ≈ 250 kJ/mol added; η split into $\eta_a bs$ and f_{ox}; estimate table added. I labelled as mean; peak = 2×. Pressure compared with literature; gauge/absolute flagged. Focus-sign conflict flagged. Supagas 12-pack data added. Machine-documentation gap recorded.

14. Sources and verification status

Source	Used for	Status
Powell, Petring, Kumar, Al-Mashikhi, Kaplan, Voisey, *J. Phys. D* 42 (2009) 015504. https://doi.org/10.1088/0022-3727/42/1/015504	ΔH ≈ 250 kJ/mol; "half the iron oxidised"; purity –3% → –50% speed; pressure ranges; iron density	**Verified** (full PDF read). CO_2 data.
Supagas, *Oxygen* technical sheet (2018 version)	12-pack volume, water capacity, grade purities	**Verified** (PDF read). Pack pressure not stated.
Supagas, *Specialty gases brochure* (2025)	Grade list, 12-pack listed	Verified (excerpt).
Sobih, Crouse & Li, *J. Phys. D* 40 (2007) 6908. DOI 10.1088/0022-3727/40/22/009	Striation-free critical speed	**Not verified.** Publisher page blocked by bot check. Scope EN43, 1–2 mm, 1 kW.
Genna, Menna, Rubino, Tagliaferri, *Appl. Sci.* 10 (2020) 4956. DOI 10.3390/app10144956	O_2 pressure 0.2–0.6 bar (CO_2)	Verified (abstract/excerpt). Full text not read.
KEM 442.81 (2010), *Role of Combustion Energy in Laser Cutting of Austenitic Stainless Steel*	40–70% combustion share; η = 50% as oxidised fraction	Excerpt only; secondary for the 40–70% range.
Laser Focus World (2010), *Cutting thick section steel with fiber lasers*	Fibre thick mild steel: 1.0 bar O_2, focus –7 mm	Excerpt only. Different machine.
Manca et al., COMSOL Users Conference (2007); arXiv 2202.06501	Pe = $v \cdot a/(2\kappa)$ for moving Gaussian source	Excerpt only.
RP Photonics, *Rayleigh length*; Kogelnik & Li (1966); ISO 11146	$z_R = \pi w_0^2/(M^2\lambda)$; M^2 definition	Verified (excerpt).
Columbia NSF workshop, *Thermal aspects in laser materials processing*	Striation vs cutting speed ("steady cutting" above ~2 m/min, thin CO_2)	Excerpt only.
Chen & Yao (2000), *High Temp. Material Processes* 4, 227	Reaction energy ≈ half at low speed	Excerpt only.
Ivarson et al. (1993); Powell et al. (1992)	Purity effect	Cited inside Powell (2009); not read directly.
8 Star Metal Engineering, KL-90010-T page	Machine envelope	Verified (page read). No laser power or gas spec.
FSCUT TubePro Laser Cutting Control System User Manual (66 pp), https://d.fscut.com/wordpress-fscut/2020/10/TubePro-User-Manual.pdf	Controller parameters: Cut focus, Cut PWR, gas DA and pressure	Verified (PDF read). No focus sign convention; pressure from DA output. May not be the same controller as the machine.
`docs/machine-kl90010t.md` (repo)	Envelope and nesting limits	Verified against the 8 Star page. Contains no laser power, gas or focus spec.

Cut-regime data spec — definitions & collection

Companion to `docs/cut-regime-models.md` (equations and trust tiers). This file defines **what we record and what each label means**, so that every run is comparable and the maps can be built from it. Data lives in `data/cut-regime/runs.csv`.

Status: **v0.4** (2026-11-09). Thresholds marked `TBD` are set from a calibration set, not guessed. Change a definition only by editing this file and noting the change in §12.

---

1. Scope and fixed series

One **series** = one fixed combination of:

- optics (lens focal length, `optics_id`), nozzle (`nozzle_id`, diameter `d_n`), gas grade and purity,
- material grade and nominal thickness `t_nom`,
- feature type (`web` / `flange` / `wall`), and face (see §5),
- focus offset convention (§3.3).

Changing any of these starts a new series. Do not pool series in the same map.

Data rows are **one sample per line** (one straight cut sample, §4.1). Corners and curves are separate feature classes (§7, §8) and are never averaged into straight-line data.

---

2. Units and conventions

Quantity	Unit	Note
Power `P_W`	W	Measured/calibrated watts at the workpiece. Log `P_set_pct` as well.
Speed `v_m_min`	m/min	Shop unit. Convert to mm/s only in formulas.
Thickness `t`	mm	**Measured** with calliper at the cut, not catalogue. `t_nom` is the catalogue value.
Gas pressure `gas_p_bar`	bar	At the head. **State gauge (barg) or absolute (bar abs)** in the series header. Shop values are unstated; see models §5.
Focus `focus_f_mm`	mm	Physical: **negative = below the top surface** (shop practice: inside or past the material), positive = above. The controller/CAM value is recorded separately in `cam_focus_mm`.
Standoff `standoff_h_mm`	mm	Nozzle tip to surface.
Energy S_{beam}	J/mm²	$S_{\mathrm{beam}} = 0.06\,\dfrac{P_W}{v_{\mathrm{m/min}}\,t_{\mathrm{mm}}}$. **Beam-only** — default S.
Energy S_{eff}	J/mm²	Only if oxidation energy is estimated (models §2: $P_{\mathrm{eff}} = \eta_{\mathrm{abs}} P + P_{\mathrm{ox}}$). Leave blank otherwise.

Rule: every table, plot and comparison states whether S is `S_beam` or `S_eff`. Default is `S_beam`.

Sign and direction: travel direction is always the direction of the beam path on the machine (see §7). Record it, do not infer it later.

---

3. Settings that must be recorded

3.1 Per run (in `runs.csv`)

`P_set_pct`, `P_W`, `v_m_min`, `focus_f_mm`, `gas_p_bar`, `standoff_h_mm`, `gas_purity_pct`.

3.2 Per series (in the series header, §10)

`optics_id`, `nozzle_id`, `nozzle_dn_mm`, `gas_grade`, `material_grade`, `t_nom_mm`, `section`, `feature`, machine id, date, operator.

3.3 Focus convention

Focus is measured from the **top surface** of the workpiece at the cut location. Physical sign: **negative = focus below the top surface** (shop practice: inside or past the material), positive = above. The controller/CAM value is logged separately in `cam_focus_mm`. If the height sensor reads from the web or a recessed channel, note it: it changes the effective focus (see `cut-regime-models.md` §8).

---

4. Samples and windows

4.1 Straight-cut sample

- A straight segment of at least **40 mm** long, steady-state.
- **Sample window:** the middle **20 mm**, excluding the first and last **10 mm** (run-in and run-out). Record `cut_length_mm` and `sample_window_mm`.
- The window must be on a single face and feature. No corner, no fillet, no hole in the window.

4.2 One face at a time

A UB/T/PFC web and flanges are separate samples. Each face (`kept` / `scrap`, §7) is graded separately.

4.3 Don't judge in the window alone

Record the whole cut (dross, bridging, separation) but grade the edge in the window.

---

5. Cut outcome — `through_code` (one per sample, mutually exclusive)

Code	Name	Criterion (apply in order, take the first that matches)
N0	No penetration	No light passes through the kerf over the window.
N1	Through, bridged	Full-thickness kerf (light passes) but the piece is still joined by melt, slag or a bridge.
N2	Forced separation	Separates only with hand force, a tap, or a hammer.
N3	Free separation	Releases by gravity or minimal handling; **no tool, no hammer, no tearing**.

Decision rules:

- **"Cut"** in the PDF sense (penetration boundary, `S_min`) = **N1 or better**, continuous through-kerf.
- **"Separated"** (usable part) = **N3**.
- **Hammering** = N2, always. Record `hit_count`.
- **Tearing a melted bridge** = N2 (`force_method = tear`). Record whether it leaves a spike or torn edge in `tear_flag`.
- **"Easily"** means N3 with a defined handling test: lift the piece off the bench, or let it drop from the clamp after release, within **5 s**. Anything needing more is N2.

- If the piece stays on the clamp or on the kerf after unclamping, it is **not** N3.

Fields: `through_code`, `force_method` (`none` / `hand` / `tap` / `hammer` / `tear`), `hit_count`, `bridge_type` (`none` / `melt` / `slag` / `mixed`), `tear_flag` (0/1).

---

6. Edge quality (apply to N2 and N3 only)

Edge quality is **several separate measurements**, not one score. ISO 9013 quality grades (thermal-cut surfaces, roughness Rz5 and perpendicularity) are the reference for comparing with industry. Verify the grade table against the standard before quoting it.

6.1 Zones

Grade every quality parameter in three zones down the thickness:

- **top** — entry, upper third
- **mid** — middle third
- **bot** — run-out, lower third

Striation commonly starts fine at the top and coarsens toward the bottom. A single average hides that.

6.2 Striation

Visual, under **10×** magnification with raking light.

Grade	Description
`S0`	None visible, even in raking light.
`S1`	Fine lines visible only in raking light.
`S2`	Lines visible in normal light.
`S3`	Deep, coarse striation, clearly visible.

Also record, for `S2/S3` only, **striation pitch** (mm, perpendicular to beam, mean of ≥ 5 lines) and **depth** (`rz_<zone>_um`, measured on the edge profile or by comparator gauge where a profilometer is not available).

6.3 Mirror

Mirror = $S0$ in **all three zones and** $rz \leq \mathrm{TBD}$ in all zones. `TBD` is set from the calibration set (§11) — do not fill it in from a published number.

`mirror_flag` = 1 only if both conditions hold. `mirror_flag` = 0 otherwise.

6.4 Dross

Dross is attached material on the **bottom (exit) edge**.

Grade	Meaning
`D0`	None.
`D1`	Loose — brushes or drops off with a fingernail.
`D2`	Adherent — needs a tool (scraper).
`D3`	Hard — cannot be removed without damage.

Record `dross_grade` (worst in the window) and `dross_height_mm` (optional, measured).

6.5 Oxide colour (heat tint)

`bright` / `straw` / `blue-brown` / `black`. Record once per sample on the **kept** face.

6.6 Taper and burr

- `taper_deg` — kerf angle from the perpendicular, measured on the kept face (°).
- `burr` — 0 none, 1 present (note position in `notes`).

6.7 Perpendicularity and dimensional check (optional, when measured)

- `perp_deg` — angularity of the cut face to the surface.
- `kerf_w_mm` — kerf width, if measurable.

---

7. Face, side and direction

- **`face`** — which profile face of the beam is cut: `web` (`v/h` in DSTV terms), `flange` (`o/u`), or `wall` (tube/SHS side). Follow `docs/canonical-orientation.md` §3 for the map to DSTV letters.
- **`kept` / `scrap`** — the face is on the piece we keep, or on the piece we throw away. The two faces of one kerf differ, so log which one is graded.
- **`direction`** — travel direction on the machine: `+x`, `-x`, `+y`, `-y` (or `CW` / `CCW` for curves). Always log, never infer.

---

8. Curves and corners

8.1 Curves

- `curve_side` — `inside` or `outside` of the curve, relative to the **kept** piece.
- `curve_radius_mm` — radius of the curve at the sample window.
- Curves are a **separate class** (`feature_class = curve`). Grade them with the same rubric, compared with a straight baseline cut in the same series.

8.2 Corners

Corners are their own class (`feature_class = corner`). Never average them into straight data.

- `corner_type` — `convex` (outside corner of the kept piece) or `concave` (inside corner).
- `corner_angle_deg` — 90 for most sections; record actual.
- `corner_radius_mm` — 0 if sharp.
- `approach_v_m_min` — speed before the corner.
- `dwell_ms` — any programmed dwell or power ramp at the corner (0 if none).
- Grade within **±5 mm** of the corner vertex and compare with the straight baseline cut in the same series.

---

9. Feature class (required on every row)

`feature_class` ∈ `straight` · `curve` · `corner` · `hole` · `end` (end-cut or run-out only). Only `straight` rows feed the first map set. Others feed later maps (§12).

---

10. Series header

Series fields are **repeated on every row** of `runs.csv` (flat CSV, no header blocks), so each row is self-describing. Fields: `series_id`, `machine`, `optics_id`, `nozzle_id`, `nozzle_dn_mm`, `gas_grade`, `gas_purity_pct`, `material_grade`, `section`, `t_nom_mm`, `feature`, `date`, `operator`. Fill `TBD` where a value is not yet known; do not guess.

Column order is fixed; the full header is the first line of `data/cut-regime/runs.csv`. Blank = not measured or not applicable. Never put commas inside a field (use `;`).

10a. Pilot series `S01_150UB14_web` (planned, not yet run)

Nine planned rows in `runs.csv` (R01–R09): five speeds at 3000 W (S sweep, R01/R03/R05/R06/R07), two power checks at v = 2.8 m/min (R02 at 2400 W, R04 at 3600 W), and three runs at the centre point (R06, R08, R09: 25%, 3000 W, 2.8 m/min). R06 is the baseline; R08 and R09 are repeats for run-to-run and grader repeatability. Planned S_beam values are in each row's `notes`. Run in a random order (not R01 to R09) and record actual execution order in `date/notes`.

Before running: confirm the % → W mapping (the pilot assumes 25% = 3000 W) and fill the TBD fields.

---

11. Calibration set (to set `TBD` thresholds)

Before the first map:

1. Cut 5 samples spanning from clearly striated to clearly mirror, on one series.
2. Grade them blind (another person scores `S` grades and `through_code`).
3. Measure `rz` on each zone with the same instrument.
4. Set the mirror `rz` threshold at the value that best separates `S0` from `S1/S2`. Record the chosen value and the date in §12.

If two graders disagree on `S` grade by more than one step, retrain on the reference set before collecting.

---

12. Change log

Date	Version	Change
2026-11-09	v0.1	First draft. Outcome codes N0–N3, striation S0–S3, dross D0–D3, zones, curve/corner classes. Mirror `rz` threshold TBD.
2026-11-09	v0.2	Pilot centre-point count corrected (three runs). Paper sheet moved to `cut-regime-data-sheets.md`.
2026-11-09	v0.4	Focus sign set to physical (negative = below top surface, shop practice). `cam_focus_mm` column added to `runs.csv`.
2026-11-09	v0.3	Gas pressure gauge/absolute stated. Sheet 0 gains gas grade, pack ID and pack pressure, and a focus-sign check.

Appendix A — Paper sheets

Moved to `docs/cut-regime-data-sheets.md` (Sheets 0–4). That file is the only printable form, so the spec and the sheets cannot drift apart.

Cut-regime data collection sheets

Printable sheets for the cut-regime test programme. Definitions and code meanings are in `docs/cut-regime-data-spec.md`. This file is only the forms.

Use one **Sheet 0** per series, one **Sheet 1** line per sample, and one **Sheet 2** grading card per sample. **Sheet 3** is the calibration set, once per series.

Transcribe to `data/cut-regime/runs.csv` at the end of each shift. Keep the paper originals.

Sheet 0 — Series header (one per series)

```
SERIES ID  _____________________     DATE _____________   OPERATOR _____________
MACHINE    KL-90010-T              SHIFT / AMBIENT TEMP _____________

OPTICS (lens focal length)  _____________   optics_id  _____________
NOZZLE  ID _____________   dn ______ mm   standoff reference: tip to surface
FOCUS SIGN CHECK  physical: NEGATIVE = below top surface (shop practice), POSITIVE = above
                  controller value entered ______   controller sign setting ______ (repo
assumes -1)
                  test: cut a focus pair (e.g. -2 and +2 on one series) and record which gives
the cut we expect
GAS     supplier Supagas   pack type [ ] 12-pack  [ ] other _________
        grade [ ] Industrial >99.5%  [ ] HP >99.8%  [ ] other _________   purity (cert) ______
%
        pack ID _____________   pack pressure at start ______ bar   at end ______ bar
        head pressure reading  [ ] gauge (barg)  [ ] absolute (bar abs)   value ______ bar

MATERIAL  grade _____________   t_nom ______ mm (catalogue)
SECTION   _____________ (e.g. 150UB14)
FEATURE   [ ] web   [ ] flange   [ ] wall (tube/SHS)   [ ] other _________
FACE      [ ] kept side graded   [ ] scrap side graded

FOCUS REFERENCE: top surface at cut location.   Height sensor source _____________
% -> kW CALIBRATION: date _________  method _____________  verified [ ] Y  [ ] N
```

Sheet 1 — Run log (one line per sample)

Fill in the run order as executed, not as planned. Put "—" where a field is not applicable. Do not leave blanks silently.

Run	Time	P%	P_W	v m/min	f mm	p bar	h mm	t meas mm	Face	Dir	Class	Cut len mm	Through N0–N3	Force	Hits	Bridge	Tear Y/N

Codes:

- **Face:** W (web), F (flange), L (wall/tube side). Add K (kept) or S (scrap).
- **Dir:** +x / −x / +y / −y (or CW / CCW for curves).
- **Class:** ST straight · CV curve · CN corner · HO hole · EN end.
- **Force:** none / hand / tap / hammer / tear.

- **Bridge:** none / melt / slag / mixed.

Handling test (N3): lift the piece off the bench, or let it drop from the clamp after release, within **5 s**, with no tool. Anything else is N2.

Sheet 2 — Grading card (one per sample, kept face unless stated)

```
RUN ________   SERIES ________   FACE  [ ] kept  [ ] scrap      CLASS ________

WINDOW: middle 20 mm of straight cut (excluding first and last 10 mm)
        [ ] corner / curve sample: distance from vertex ______ mm

PENETRATION & SEPARATION
  Through code:   N0   N1   N2   N3        Force:  none / hand / tap / hammer / tear
  Hits: ______     Bridge: none / melt / slag / mixed     Tear spike:  Y / N

STRIATION (raking light, 10x)      top        mid        bot
  Grade S0 / S1 / S2 / S3          ____       ____       ____
  Pitch (mm, S2/S3 only)           ____       ____       ____
  Depth Rz (um)                    ____       ____       ____

MIRROR
  All zones S0 and Rz < TBD?       [ ] Y   [ ] N  (TBD = ______ um, from calibration)

DROSS (bottom edge, worst in window)
  D0 none   D1 loose   D2 adherent   D3 hard        Height (mm, optional) ______

OXIDE (kept face)     [ ] bright   [ ] straw   [ ] blue-brown   [ ] black

TAPER ______ deg     BURR  [ ] none   [ ] present (position ________)
PERP ______ deg (optional)     KERF W ______ mm (optional)

PHOTO REF _______________     GRADED BY _______________   BLIND TO SETTINGS  [ ] Y  [ ] N
NOTES ____________________________________________________________________________
```

Grading rules:

- Grade the **window** only, but record dross and bridging for the whole cut.
- Use **raking light at 10×** for striation. Check all three zones.
- A grader who knows the settings must say so. Blind grading is preferred for calibration.
- If two graders differ on the striation grade by more than one step, re-check before logging.

Sheet 2a — Curve and corner add-on

Fill in only when the sample is `CV` or `CN`. Attach to the Sheet 2 card.

```
CLASS:   CV curve   [ ] inside   [ ] outside (relative to kept piece)   radius ______ mm
         CN corner  [ ] convex   [ ] concave   angle ______ deg   radius ______ mm

APPROACH SPEED ______ m/min    DWELL / RAMP ______ ms (0 if none)
DIRECTION  [ ] CW   [ ] CCW   [ ] other ________

GRADE within +-5 mm of vertex:   top ____  mid ____  bot ____
COMPARE with straight baseline cut on the same series:  [ ] better  [ ] same  [ ] worse
```

Sheet 3 — Calibration set (once per series, before the map)

Purpose: set the mirror Rz threshold (spec §11). Use five samples that span striated to mirror on one series.

```
CAL SERIES _______________    DATE _____________    GRADER A ______   GRADER B ______

SAMPLE   Set (P_W, v)   Grader A striation (top/mid/bot)   Grader B striation   Rz top/mid/bot
(um)
C1   _____________        ______ / ______ / ______       ______ / ______ / ______
____/____/____
C2   _____________        ______ / ______ / ______       ______ / ______ / ______
____/____/____
C3   _____________        ______ / ______ / ______       ______ / ______ / ______
____/____/____
C4   _____________        ______ / ______ / ______       ______ / ______ / ______
____/____/____
C5   _____________        ______ / ______ / ______       ______ / ______ / ______
____/____/____

Agreement: grades within one step?    [ ] Y   [ ] N  (N = retrain before collecting)
Rz instrument _____________   calibrated date _____________
Chosen mirror Rz threshold: ______ um      Rationale (S0 vs S1/S2 separation) _____________
Date of decision _____________   Signed _____________
```

Sheet 4 — Pilot checklist (S01_150UB14_web)

Run in random order. Tick when done.

```
[ ] Sheet 0 complete; TBD fields filled (optics, nozzle, gas, focus, standoff, gas p)
[ ] $\%$ $\rightarrow$ kW confirmed (measure, do not assume 25% = 3000 W)
[ ] t_meas measured on the part, not catalogue
[ ] Sample length $\ge 40$ mm, window clear of corners, holes and fillets
[ ] Run order randomised: ________________________________________
[ ] R01  [ ] R02  [ ] R03  [ ] R04  [ ] R05  [ ] R06 (centre)  [ ] R07  [ ] R08 (centre)  [ ]
R09 (centre)
[ ] Each cut graded blind on Sheet 2 by two graders for at least the centre points
[ ] Photos taken before handling (photo ref recorded)
[ ] Transcribed to runs.csv
```

Quick code reference

Field	Codes
`through_code`	N0 no penetration · N1 through, bridged · N2 forced separation · N3 free separation
`force_method`	none · hand · tap · hammer · tear
`bridge_type`	none · melt · slag · mixed
`striation`	S0 none · S1 raking light only · S2 normal light · S3 deep, coarse
`dross_grade`	D0 none · D1 loose · D2 adherent · D3 hard
`oxide_colour`	bright · straw · blue-brown · black
`feature_class`	straight · curve · corner · hole · end
`face`	web · flange · wall
`kept_scrap`	kept · scrap